

To get the temperature from an NTC 3950 thermistor, you'll need to measure its resistance and then use the Steinhart-Hart equation to convert the resistance to temperature. Here's a breakdown of the process:

1. **1. Measure Resistance:**

Connect the thermistor to a circuit that allows you to measure its resistance. This can be done using a digital multimeter or by using a voltage divider and reading the voltage across the thermistor.

2. **2. Use the Steinhart-Hart Equation:**

The Steinhart-Hart equation is a mathematical model that describes the relationship between the resistance and temperature of an NTC thermistor. The equation is:

Código

$$1/T = 1/T_0 + (1/\beta) * \ln(R / R_0)$$

Where:

- T is the temperature in Kelvin.
- T₀ is the reference temperature (typically 25°C or 298.15 K).
- R is the resistance of the thermistor at the measured temperature.
- R₀ is the resistance of the thermistor at the reference temperature (typically 3950 ohms for an NTC 3950).
- β is the material-specific constant for the thermistor, often provided on the thermistor's datasheet or in the documentation. For an NTC 3950, the β value is around 3950.

1. **Calculate Temperature:** Once you have the measured resistance (R) and the β value (typically 3950 for an NTC 3950), you can plug these values into the Steinhart-Hart equation and solve for T (temperature in Kelvin). Convert the temperature from Kelvin to Celsius by subtracting 273.15.

Example:

Let's say you measure the resistance of the NTC 3950 to be 5000 ohms. You can use the following steps:

1. **Reference Values:** R₀ = 3950 ohms, T₀ = 298.15 K, β = 3950
2. **Plug into Equation:**

Código

$$1/T = 1/298.15 + (1/3950) * \ln(5000 / 3950)$$

1. **Solve for T:** Calculate the temperature T in Kelvin. Then, convert to Celsius.

Important Considerations:

- **Accuracy:**

The accuracy of your temperature measurement will depend on the accuracy of your resistance measurement and the accuracy of the β value used in the calculation. If you need high accuracy, consider using a calibrated NTC thermistor and calibrating the β value for your specific sensor.

- **Temperature Range:**

The Steinhart-Hart equation is generally accurate for a wide range of temperatures, but its accuracy may vary depending on the specific NTC thermistor and the temperature range.

- **Calibration:**

For critical applications, it's recommended to calibrate the thermistor against a known reference temperature to ensure accurate temperature readings, [says a Duet3D forum post](#).